AutoFuzzer-AI: Simple Project Report

1. Project Idea

This project builds an intelligent firmware testing system using an ESP32 and an STM32. The ESP32 automatically sends normal and malformed communication packets through UART, SPI, I^2C or CAN to test whether the STM32 firmware can handle unexpected inputs safely.

2. Hardware

ESP32: Generates test packets, runs AI models, monitors heartbeat.
STM32: Runs the firmware being tested and periodically sends a heartbeat signal.

3. AI Features

Adaptive Packet Mutation: The AI learns which packet mutations are most likely to reveal firmware bugs and prioritizes those mutations.

Crash Prediction: A TinyML model analyzes heartbeat timing, packet length, mutation type and communication speed to estimate crash probability before the firmware actually crashes.

4. Workflow

1. ESP32 creates a packet.
2. AI chooses a mutation.
3. Packet is sent to STM32.
4. STM32 processes the packet and sends heartbeat.
5. AI predicts instability.
6. If a crash occurs, the ESP32 logs the packet and displays the result.

5. Expected Output

The system reports the protocol, mutation type, crash probability, heartbeat status and whether the firmware passed or failed. This helps developers find bugs faster and improve firmware reliability.

6. Advantages

- Low-cost hardware.
- Automated firmware robustness testing.
- AI-guided smarter fuzzing.
- Predictive crash detection.
- Reusable for multiple communication protocols.